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COURSE FEEDBACK AND RATING SYSTEM

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EduFeedback

ABSTRACT

The Feedback system is a web-based application designed to replace traditional paper-based feedback methods. It allows students to submit feedback easily and securely, with options for anonymity. The system ensures fair data collection by preventing duplicate submissions and provides useful insights through an admin dashboard. It improves efficiency, transparency, and decision-making in educational institutions.

PROBLEM STATEMENT

- Traditional paper-based feedback systems are slow, hard to analyze, prone to data loss, and offer no real-time insights. Faculty and administrators cannot act quickly on feedback. Students often avoid honest feedback due to fear of identification.

PROBLEM SOLUTION

- EduFeedback is a modern, full-stack web application that digitizes and enhances the traditional paper-based course feedback process in educational institutions. It replaces manual feedback forms with a secure, interactive, and data-driven digital platform that benefits both students and administrators.
- EduFeedback solves this by providing a role-based digital platform where students can submit rich, multi-dimensional feedback anonymously or with identity, and administrators can view real-time analytics through visual dashboards.

SYSTEM ARCHITECTURE (FRONTEND)

The frontend uses a component-based React architecture:

- Pages: Login, Register, Dashboard, SubmitFeedback, AdminPanel — Full page views
- Components: StarRating, EmojiReaction, NPSRating, ThumbsRating, NumericSlider, LikertScale, RankingQuestion, RubricScoring, IdentityModeSelector, DynamicFeedbackForm, ExitIntentDetector, Navbar, AdminRoute, ProtectedRoute
- Context: AuthContext provides global authentication state using React Context API + useAuth() hook
- Services: api.js, authService.js, courseService.js, feedbackService.js — Axios-based API calls

SYSTEM ARCHITECTURE (BACKEND)

The backend follows the MVC (Model-View-Controller) pattern:

- Models: User, Course, Lesson, Feedback, FeedbackForm, FeedbackResponse — Mongoose schemas defining data structure
- Controllers: authController, courseController, feedbackController — Contain all business logic
- Routes: Map HTTP methods + endpoints to specific controller functions
- Middleware: protect() for JWT verification, adminOnly() for role-based access
- Config: db.js handles MongoDB connection with graceful shutdown handling

SYSTEM ARCHITECTURE (DATABASE)

Collection	Key Fields	Purpose
users	_id, name, email, password (hashed), role (student/admin), timestamps	Stores all registered users with bcrypt-hashed passwords
courses	_id, courseName, courseCode, facultyName, department, semester, isActive, timestamps	Course catalogue managed by admin
feedback	_id, studentId (ref), courseId (ref), lessonId (ref), feedbackType (micro/macro), rating (1-5), teachingQuality, courseContent, overallSatisfaction, emojiReaction, thumbsUpDown, npsScore, sliderRatings, identityMode, pseudonym, comment, anonymous, timestamps	Core feedback data with multi-dimensional ratings
lessons	_id, courseId (ref), title, order, description, timestamps	Individual lessons within a course
feedbackforms	_id, courseId, title, questions (array), isActive, type, timestamps	Admin-created dynamic survey forms
feedbackresponses	_id, formId (ref), courseId (ref), studentId (ref), answers (array), identityMode, triggerSource, timestamps	Responses to dynamic forms

KEY FEATURES

- Authenication and security
- Multi-Dimensional feedback system
- Three identity modes
- feedback types : micro vs Macro
- Exit intent detection
- Admin Analytics Dashboards
- Dynamic feedback forms

FUTURE SCOPE

- AI-based sentiment analysis
- Mobile application
- Real-time notifications
- Advanced analytics

CONCLUSION

The Feedback Course Feedback and Rating System provides a modern and efficient solution to replace traditional feedback methods. By using a secure web-based platform, the system enables students to submit feedback easily and honestly, including anonymous responses. It ensures fairness by preventing duplicate submissions and offers multi-dimensional insights through an admin dashboard. This helps institutions analyze feedback quickly and take necessary actions to improve teaching quality. Overall, the project enhances transparency, saves time, and makes the feedback process more reliable and effective.